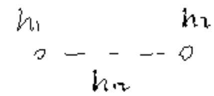
Renormalization Group

§ General Introduction

RG：一种简化方式

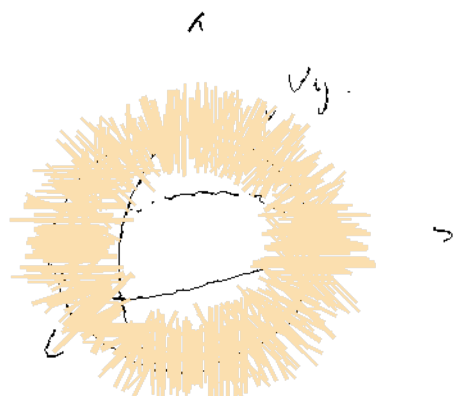
两体问题：$H = h_1 + h_2 + h_{12}$



$\Rightarrow H_{eff}^{(1)} = h_1 + \Sigma$　　$\Sigma \sim Tr_2(h_{12} + h_2)$

只有1的自由度　　correction

费米球，相互作用



Ω_1 : $|\vec{k}| \in [k_F, k_F + \delta\Lambda]$

$H = H_{\Omega_1} \oplus H_{\Omega_2}$

$\Rightarrow H_{eff} = H_{\Omega_1} + \Sigma$　　$\Sigma \sim Tr_{\Omega_2}(H)$

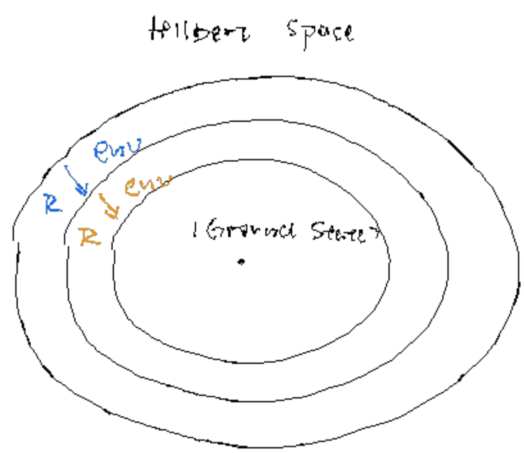
"Group"：$H_0 \xrightarrow{R} H_{eff_1} \xrightarrow{R} H_{eff_2}$ —— ...　　$b > 1+\epsilon$

$\Lambda_0 \to \Lambda_1 = \frac{\Lambda_0}{b} \to \Lambda_2 = \frac{\Lambda_1}{b}$

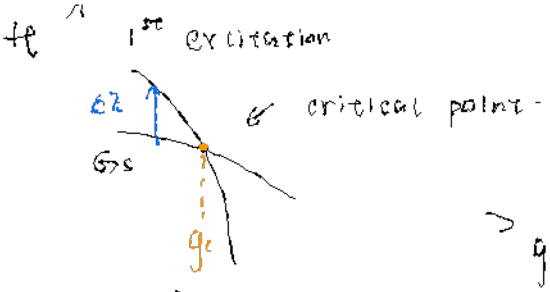
Ising Model : $D = 2^N$

High Energy State : Local Interaction

↳ R of High E : "Mean Pooling" in Real Space



Phase transition: $H(g) = H_0 + gH'$



$\delta E \propto \delta k \propto \frac{1}{\delta x}$

$g = g_c$: 关联长度 $\xi \sim \delta x \to \infty$

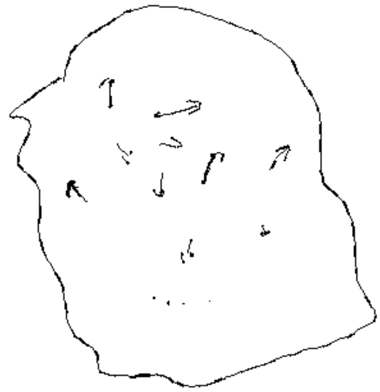
$\Rightarrow$ 自相似

$$H^{(0)} \xrightarrow{\hat{R}} H_{eff}^{(1)} \xrightarrow{\hat{R}} H_{eff}^{(2)} \cdots H_{eff}^{(N)}$$

$$\xi^{(0)} \qquad \xi^{(1)} = \xi^{(0)}/b \qquad \xi^{(2)} = \xi^{(0)}/b^2 \qquad \infty/b = \infty$$

a fixed point ?

§ Statistical Field Theory.



Landau Phase Transition

Order Parameter : $\vec{m} \to \vec{m}(\vec{x})$

Free Energy : $F(m) = a_0 + a_2 m^2 + \cdots$

$\to F[\vec{m}(\vec{x})]$

$$\begin{cases} C(T) \sim |T - T_c|^{-\alpha} \\ m(T) \sim (T_c - T)^{\beta} \\ \chi(T) = \left(\frac{\partial m}{\partial B}\right)_T \sim |T - T_c|^{-\gamma} \\ m(B) \sim B^{1/\delta} \end{cases}$$

$G^{(2)}(\vec{x}) := \langle \vec{m}(\vec{0})\, \vec{m}(\vec{x}) \rangle$ $G_c^{(2)}$: 涨落关联.

$G_c^{(2)}(\vec{x}) := \langle \vec{\phi}(\vec{0})\, \vec{\phi}(\vec{x}) \rangle \sim \begin{cases} x^{-(d-2+\eta)} & d:\ \text{system dim near critical point.} \\ e^{-x/\xi} & \xi:\ \text{correlation length.} \end{cases}$

$\vec{\phi}(\vec{x}) = \vec{m}(\vec{x}) - \langle \vec{m}(\vec{x}) \rangle$

$\xi \sim |T - T_c|^{-\nu}$

§ RG Formalism

H : microscopic $\to$ $H_{eff} = H_{eff}[\vec{m}(\vec{x})]$

$$Z = \mathrm{Tr}\,(e^{-\beta H}) = \sum_{\{s\}} e^{-\beta E_{\{s\}}}$$

$$\to \quad = \sum_{\{\vec{m}(\vec{x})\}} e^{-\beta H_{eff}} \qquad H_{eff} = \int d^d x\ \Phi_{eff}[\vec{m}(\vec{x})]$$

$$= \int e^{-\beta H_{eff}}\, \mathcal{D}[\vec{m}(\vec{x})]$$

$$= \int e^{-\beta \int d^d x\ \Phi_{eff}[\vec{m}(\vec{x})]}\, \mathcal{D}[\vec{m}(\vec{x})]$$

From Spin Rotation Symmetry :

$$\beta H_{eff} = \beta F_0 + \int d^d x \left[\frac{t}{2} \vec{m}^2(\vec{x}) + u(\vec{m}\cdot\vec{m})(\vec{m}\cdot\vec{m}) + \frac{k}{2}(\nabla\cdot\vec{m}) + \cdots - \vec{h}\cdot\vec{m}(\vec{x}) \right]$$

$$G^c_{\alpha\beta}(\vec{x}) = \langle (\vec{m}(\vec{x}) - \langle m\rangle)(\vec{m}(\vec{x}') - \langle m\rangle) \rangle \qquad \langle m\rangle = \frac{1}{Z}\,Tr(e^{-\beta H} m)$$

$$= \frac{1}{V}\sum_{q,q'} e^{i(\vec{q}\cdot\vec{x} + \vec{q}'\cdot\vec{x}')} \phi_{\alpha\vec{q}}\,\phi_{\beta\vec{q}'} \qquad = \frac{1}{Z}\int e^{-\beta H_{eff}} D[\vec{m}(\vec{x})]$$

Fourier Transformation: $\vec{m}(\vec{x}) = \frac{1}{\sqrt{V}}\sum_q \vec{m}_{\vec{q}}\, e^{i\vec{q}\cdot\vec{x}}$

$\vec{m} = \bar{m} + \phi$

$$\beta H_{eff} = \beta F_0 + \int d^d x \left(\frac{t}{2}\bar{m}^2 + u\bar{m}^4\right) + \int d^d x \left[\frac{k}{2}(\nabla\cdot\phi)^2 + \frac{t^2 + 4u\bar{m}\bar{m}}{2}\phi^2\right]$$

$$= \beta F_0 + V\cdot\left(\frac{t}{2}\bar{m}^2 + u\bar{m}^4\right) + \int d^d x \left[\frac{k}{2}(\nabla\cdot\phi)^2 + \frac{t^2 + 4u\bar{m}\bar{m}}{2}\phi^2\right]$$

$$= C + \int \frac{d^d q}{(2\pi)^d}\left(\frac{k}{2}q^2 + \frac{t^2 + 4u\bar{m}\bar{m}}{2}\right)\phi_{\alpha\vec{q}}\,\phi_{\alpha\vec{q}} \qquad \nabla\phi \to ?$$

$$\xi^{-2} = \frac{t^2 + 4u\bar{m}\bar{m}}{k}$$

$$= C + \int \frac{d^d q}{(2\pi)^d}\left(\frac{k}{2}q^2 + \frac{k}{2}\xi^{-2}\right)\phi_{\alpha\vec{q}}\,\phi_{\alpha\vec{q}}$$

$$\langle \phi_{\alpha q}, \phi_{\beta q'} \rangle = \int D\phi_{\alpha\vec{q}}\; e^{-\beta H_{eff}}\; \phi_{\alpha\vec{q}}\,\phi_{\beta\vec{q}'}$$

$$= \int D\phi_{\alpha\vec{q}}\; e^{-\left[C + \int \frac{d^d q}{(2\pi)^d}\frac{k}{2}(q^2 + \xi^{-2})\phi_{\alpha\vec{q}}\,\phi_{\alpha\vec{q}}\right]}\; \phi_{\alpha\vec{q}}\,\phi_{\beta\vec{q}'}$$

$$= \frac{\delta_{\alpha\beta}\,\delta_{q,-q'}}{k^2(q^2 + \xi^{-2})} \propto \begin{cases} e^{-|x|/\xi} & |x| \gg \xi \\ \underbrace{|x|^{2-d}}_{d=2} & |x| \ll \xi \end{cases}$$

$$\phi(\vec{x}) = |\phi(\vec{x})|\, e^{i\theta(x)}$$

...

$r < \xi$, $G^{(c)}(r) \sim r^{-(d-2+\eta)}$

At Critical Point : $\xi \to \infty$ $\Rightarrow$ $G^{(c)}(r) \sim \cdots$ is universal.

$r \to r' = \lambda r$

$G^{(c)} \to G^{(c)'} = \lambda^{?} G^{(c)}$ self - similarity.

1° coarse - graining

$\vec{m}(\vec{x}) = \frac{1}{b^d} \int_{\vec{x} \in V} d\vec{x}' \, \vec{m}(\vec{x}')$ V : centered at $\vec{x}$

2° rescaling $\vec{x}_{new} = \frac{\vec{x}_{old}}{b}$ $(b > 1)$ $d_{new} = b \, d_{old}$.

3° renormalization $\vec{m}_{new} = \frac{\vec{m}_{old}}{z}$

1° + 2° + 3° : $\vec{m}_{new}(\vec{x}_{new}) = \frac{1}{z b^d} \int_{x \in V} d^d \vec{x}' \, \vec{m}_{old}(\vec{x}')$ V centered at $b\vec{x}_{new}$

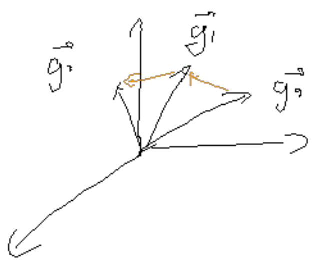
$Z = \int D\vec{m}_{old}(\vec{x}_{old}) \, e^{-\beta H_{old}[\vec{m}_{old}(\vec{x}_{old})]} \simeq \int D\vec{m}_{new}(\vec{x}_{new}) \, e^{-\beta H_{new}[\vec{m}_{new}(\vec{x}_{new})]}$

RG

$H_{old}[\vec{m}_{old}(\vec{x}), \vec{g}] \Rightarrow H_{new}[\vec{m}_{new}(\vec{x}_{new}), \vec{g}']$

$\vec{g}' = \vec{g}'(\vec{g})$

$\vec{g}_0 \overset{RG}{\Rightarrow} \vec{g}_1 \overset{RG}{\Rightarrow} \cdots$



Let $b = 1 + \varepsilon$, $\varepsilon \to 0$

$= 1 + dl$

RG Flow Equation : $\frac{d\vec{g}}{dl} = \beta(\vec{g})$

$$\beta \mathcal{H} = \int d^d x \ \frac{t}{2} m^2 + \cdots + (-\vec{h} \cdot \vec{m})$$

$$\vec{g} = (t, \vec{h})$$

$$\begin{tikzcd} \mathcal{H}_0 \arrow[r, bend left, "b_1"] \arrow[rr, bend right, "b_1 b_2"'] & \mathcal{H}_1 \arrow[r, bend left, "b_2"] & \mathcal{H}_2 \end{tikzcd}$$

RG : $b = 1 + dl = e^{dl}$

$$\vec{g}_b = \vec{g}_b(\vec{g}) \quad \begin{cases} t_b = t_b(t) \\ \vec{h}_b = \vec{h}_b(\vec{h}) \end{cases} \qquad \text{Group} \begin{cases} t_{b_1} t_{b_2} = t_{b_1 \cdot b_2} \\ h_{b_1} h_{b_2} = h_{b_1 \cdot b_2} \end{cases}$$

$$\Rightarrow \begin{cases} t_b = t\, b^{y_t} = t(1+dl)^{y_t} = t(1 + y_t\, dl) \\ \vec{h}_b = \vec{h}\, b^{y_h} = \vec{h}(1+dl)^{y_h} = \vec{h}(1 + y_h\, dl) \end{cases}$$

$$\Rightarrow \quad \frac{dt}{dl} = y_t\, t \quad , \quad \frac{d\vec{h}}{dl} = y_h \vec{h}$$

$$\begin{cases} y > 1 & : \text{relevant} \\ y < 1 & : \text{irrelevant} \end{cases}$$

Fixed Point : $\vec{g} = \vec{g}^*$ s.t. $\frac{d\vec{g}}{dl} = \beta(\vec{g}) = 0$

§ Momentenum Space RG

$$\beta H = \int d^d x \left\{ \frac{t}{2} m^2 + \frac{K}{2} (\nabla m)^2 + \frac{L}{2} (\nabla^2 m)^2 - \vec{h} \cdot \vec{m} \right)$$

$\downarrow$ Fourier Transformation.

$$\beta H = \int_0^\Lambda \frac{d^d q}{(2\pi)^d} \left(\frac{t + Kq^2 + Lq^4}{2} |m_{\vec{q}}|^2 - \vec{h} \cdot \vec{m} \right)$$

Lattice Cutoff: $\Delta x \sim a$, $\Delta q \sim \Lambda = \frac{1}{a}$

$$\vec{m}_{\vec{q}} = \vec{m}_{|\vec{q}| < \frac{\Lambda}{b}} \oplus \vec{m}_{\frac{\Lambda}{b} < |\vec{q}| < \Lambda}$$ = slow mode $\oplus$ fast mode.

$$= \tilde{\vec{m}} \oplus \vec{\sigma}$$

$$Z = \int \mathcal{D} \vec{m}\, e^{-\beta H [\tilde{\vec{m}} \oplus \vec{\sigma}]}$$ 只有二次项时，没有耦合。

$$= \int \mathcal{D} \tilde{\vec{m}}\, e^{-\beta H [\tilde{\vec{m}}]} \mathcal{D} \vec{\sigma}\, e^{-\beta H [\vec{\sigma}]}$$

$$= C \int \mathcal{D} \tilde{\vec{m}}\, e^{-\beta H [\tilde{\vec{m}}]}$$

$$= C \int \mathcal{D} \tilde{\vec{m}}(q)\, e^{-\int_0^{\frac{\Lambda}{b}} \frac{d^d q}{(2\pi)^d} \left[\frac{t + Kq^2 + Lq^4}{2} |\tilde{\vec{m}}_q|^2 - \vec{h} \cdot \tilde{\vec{m}}(\vec{q} = 0) \right]}$$

$\vec{x}_{new} = \frac{\vec{x}_{old}}{b}$, $\vec{q}_{new} = b\, \vec{q}_{old}$

$$= C \int \mathcal{D} \tilde{\vec{m}}(q')\, e^{-\int_0^\Lambda \frac{d^d q'}{(2\pi b)^d} \frac{t + Kq'^2 b^{-2} + Lq'^4 b^{-4}}{2} |\tilde{\vec{m}}(q')|^2 - \vec{h} \cdot \tilde{\vec{m}}(0)}$$

$\vec{m}_{new} = \frac{\vec{m}_{old}}{Z}$

$$= C \int \mathcal{D} \vec{m}'(q')\, Z\, e^{-\int_0^\Lambda \frac{d^d q'}{(2\pi b)}}$$

$$\begin{cases} t' = t Z^2 b^{-d} \\ K' = K b^{-2-d} Z^2 \\ L' = L b^{-4-d} Z^2 \end{cases}$$

match K:

$K' = K \Rightarrow Z = b^{1+\frac{d}{2}}$

RG Flow Eq.

$$\begin{cases} \frac{dt}{dl} = 2t \\ \frac{dL}{dl} = -2L \end{cases}$$

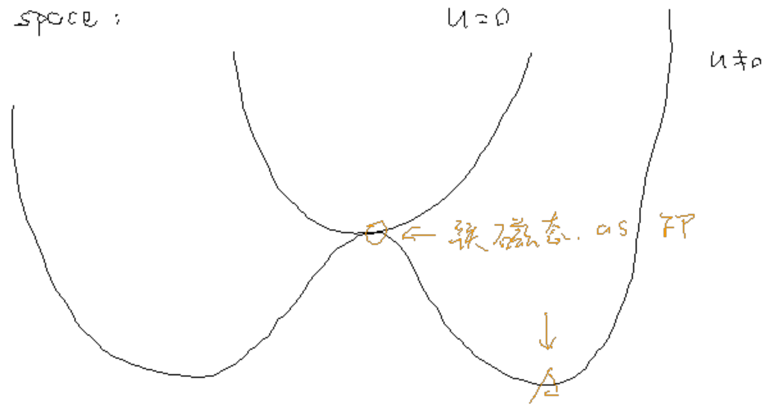
$b = 1 + dl$

$h' = hz$

$$\frac{dh}{dl} = (1 + \frac{d}{2})h$$

Fixed Point : $(t, L, h) = (0, 0, 0)$

g space :



βH 中加入 : $\int d^d x \; u m^4 = \int d^d x (\vec{m} \cdot \vec{m})(\vec{m} \cdot \vec{m})$

$$\Rightarrow \int \frac{d^d q_1 \, d^d q_2 \, d^d q_3}{(2\pi)^{3d}} [\vec{m}(\vec{q}_1) \cdot \vec{m}(\vec{q}_2)][\vec{m}(\vec{q}_3) \cdot \vec{m}(\vec{q}_4 = -\vec{q}_1 - \vec{q}_2 - \vec{q}_3)]$$

$\vec{m} = \tilde{\vec{m}} \oplus \vec{\sigma}$

$$Z = \int D\vec{m} \; e^{-\beta H_{eff}[\vec{m}]} = \int D\tilde{\vec{m}} \, D\vec{\sigma} \; e^{-\beta H_{eff}[\tilde{\vec{m}}, \vec{\sigma}]}$$

$$= \int D\tilde{\vec{m}} \, D\vec{\sigma} \; e^{-\beta H_{eff}[\tilde{\vec{m}}]} \, e^{-\beta H_{eff}[\vec{\sigma}]} \, e^{-u[\tilde{\vec{m}}, \vec{\sigma}]}$$

$$= \int D\tilde{\vec{m}} \; e^{-\beta H_{eff}[\tilde{\vec{m}}]} \cdot \int D\vec{\sigma} \; e^{-\beta H_{eff}[\vec{\sigma}]} e^{-u[\tilde{\vec{m}}, \vec{\sigma}]}$$

$$= Z_6 \int D\tilde{\vec{m}} \; e^{-\beta H_{eff}[\tilde{\vec{m}}]} \cdot \frac{1}{Z_6} \int D\vec{\sigma} \; e^{-\beta H_{eff}[\vec{\sigma}]} e^{-u[\tilde{\vec{m}}, \vec{\sigma}]}$$

$$= Z_6 \cdot \int D\tilde{\vec{m}} \; e^{-\beta H_{eff}[\tilde{\vec{m}}]} \cdot \langle e^{-u} \rangle_6 \qquad Z_6 = \int D\vec{\sigma} \; e^{-\beta H_{eff}[\vec{\sigma}]}$$

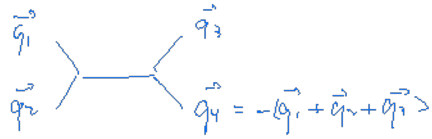
$$= Z_6 \cdot \int D\tilde{\vec{m}} \; e^{-\beta H'_{eff}[\tilde{\vec{m}}]}$$

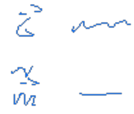
$$\hookrightarrow \beta H'_{eff}[\tilde{\vec{m}}] = \beta H_{eff}[\tilde{\vec{m}}] - \ln \langle e^{-u} \rangle_6 \qquad \ln(1 - u + \frac{1}{2}u^2 + \cdots)$$

$$\simeq \beta H_{eff}[\tilde{\vec{m}}] - \langle u \rangle_6 + \frac{1}{2} \langle u^2 \rangle_6$$

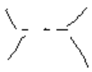
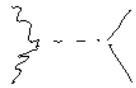
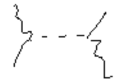
$$\langle u \rangle_6 = \frac{1}{Z_6} \int D\vec{\sigma} \; e^{-\beta H_{eff}[\vec{\sigma}]} \, u[\tilde{\vec{m}}, \vec{\sigma}]$$

Feynman Diagram

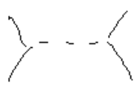


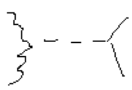


$u[\vec{m},\vec{\sigma}]$:

$1\times$  $4\times$  $= 0$ $+\ 6\times$ (4x)  / (2x) 

$+\ 4\times$  $= 0$ $+\ 1\times$  $= C$

①  : $u\int \frac{d^dq_1}{(2\pi)^d}\frac{d^dq_2}{(2\pi)^d}\frac{d^dq_3}{(2\pi)^d}\ \tilde{m}_\alpha\tilde{m}_\alpha\tilde{m}_\beta\tilde{m}_\beta$

② A  : $u\int \frac{d^dq_1\,d^dq_2\,d^dq_3}{(2\pi)^{3d}}\ \vec{\sigma}_\alpha(q_1)\,\vec{\sigma}_\alpha(q_2)\,\tilde{m}_\beta(q_3)\,\tilde{m}_\beta(-q_1-q_2-q_3)$

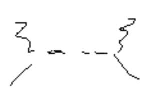
$$\langle\sigma_\alpha(q_1)\sigma_\alpha(q_2)\rangle_\sigma = \frac{1}{Z_\sigma}\int \mathcal{D}\vec{\sigma}\ e^{-\beta H[m][\vec{\sigma}]}\ \sigma_\alpha(q_1)\sigma_\alpha(q_2)$$

$$= \frac{\delta(\vec{q}_1+\vec{q}_2)\,\delta_{\alpha\alpha}}{t^2+kq_1^2+Lq_1^4}(2\pi)^d$$

$$u\int\frac{d^dq_1\,d^dq_2\,d^dq_3}{(2\pi)^{3d}}\ \frac{\delta(\vec{q}_1+\vec{q}_2)\,\delta_{\alpha\alpha}}{t^2+kq_1^2+Lq_1^4}(2\pi)^d\ \tilde{m}_\beta(\vec{q}_3)\,\tilde{m}_\beta(-q_1-q_2-q_3)$$

$\delta_{\alpha\alpha} = n$

$$= u\,n\int_0^{\Lambda/b}\frac{1}{(2\pi)^{2d}}\,d^dq_3\,|m_\beta(\vec{q}_3)|^2\int_{\Lambda/b}^{\Lambda}\frac{d^dq_1}{t^2+kq_1^2+Lq_1^4}$$

② B  : $u\int\frac{d^dq_1\,d^dq_2\,d^dq_3}{(2\pi)^{3d}}\ \tilde{\sigma}_\alpha(q_1)\,\tilde{m}_\alpha(q_2)\,\tilde{\sigma}_\beta(q_3)\,\tilde{m}_\beta(-q_1-q_2-q_3)$

!

$$= u\int_0^{\Lambda/b}\frac{d^dq}{(2\pi)^d}\frac{1}{t^2+kq^2+Lq^4}\int_{\Lambda/b}^{\Lambda}\frac{d^dq}{(2\pi)^d}\,|m_\beta(q)|^2$$

$\beta H_{eff} = \dots$

$\vec{x}' = \frac{\vec{x}}{b}$, $\vec{q}' = b\vec{q}$

$\vec{m}' = \frac{\vec{m}}{z}$

$k' = \; \cdot \; = k \Rightarrow z = z(b)$

$\frac{dm}{dl} =$

§ Real Space RG

1D Ising Model:

$$Z = \sum_{\{\sigma_i\}} e^{-\beta \sum [-J\sigma_i\sigma_{i+1} - h\sigma_i]}$$

$$Z' = \sum_{\{S_i\}} e^{-\beta H_{eff}[\{S_i\}]}$$

§ Final

2× 热力学、 2× 玻/费/最大概然 ~ 2× 系综.

① 热力学基础
- 热力学基本方程、
- S (计算)

② 均匀平衡态.
- 气体节流 ☆
- 绝热去磁 ☆
- 热辐射.

③ 相/相变、
- 相平衡判据.
- Landau 相变理论. ☆ F → 稳定性/热力学函数. (临界点表现)

④ ×

⑤ 近独立粒子系统. 能均分定理.

经典 → 量子统计
- 双原子分子 (振动)
- 固体热容 ☆

⑥ 量子统计 BEC ☆

强简并电子气 (费米气) ☆
- E_F, $D(\varepsilon)$
- finite T, $\bar{E}$, C_V?

⑦ 系综 Z → 热力学量.

涨落（三种系综）

应用 $\begin{cases} \text{非理想气体} \\ \text{相对论气体} \end{cases}$

⑧ 1D Ising Model 严格解.